Kinwise.

Family medicine-review co-pilot · private, on-device AI

UnivaBio 2026 · AI for Human Health | Fictional demo data

Video: on the Devpost page · runs in the browser, no backend

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Helps families caring for an older adult on many medicines notice what changes after a medicine change, and bring a clear question to the visit.

Live demo: albertzorzolan.github.io/UnivaBioBabson → Try as Alex **Code:** github.com/albertzorzolan/UnivaBioBabson/tree/medwatch-v2

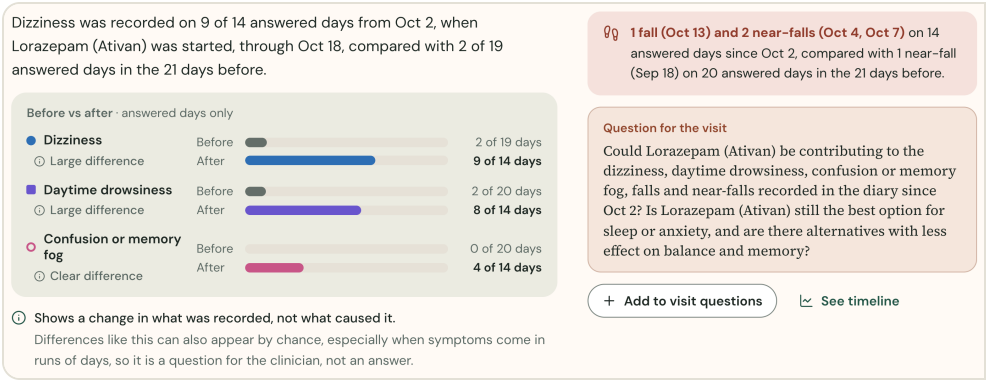
The problem

43.0% NCHS, Health, United States
of US adults 65+ took **5 or more prescription drugs** in a 30-day period (2017–Mar 2020).

37% Gandhi et al., NEJM 2003
of adverse drug events in primary care whose harm could have been reduced were attributed to **the patient not telling the doctor**.

More than 1 in 4 CDC, Facts About Older Adult Falls
older adults **fall each year**, but **less than half tell their doctor**.

What's different. Diary apps (MyTherapy, CareClinic) chart symptoms but do not document before/after counts around medicine starts; clinician tools (MedStopper, MedSafer) apply Beers/STOPP-type rules without the family's observations. Kinwise does both.[4]



What Alex brings to the visit

The caregiver Overview in the live demo (fictional data): counts on answered days only, a reminder that a change is not a cause, and a question for Margaret's clinician.

Who it is for, and where

The **family caregiver** (Alex) and the **older adult** (Margaret, 78, 8 medicines), who checks in on a large-text screen; the brief is for her **doctor or pharmacist**.

For the Medicare **Annual Wellness Visit** (whose cognitive check must consider concerns raised by family[1]) or a pharmacist's **medication review**, which may be done with the caregiver[2].

> **Why it could help:** in EMPOWER[3] (cluster RCT, 303 long-term benzodiazepine users aged 65–95), **62%** of those given a brochure raised stopping with their doctor or pharmacist, and **27% vs 5%** (brochure vs usual care) had discontinued at 6 months (the brochure included a taper plan; Kinwise leaves that to the clinician). A reason to test Kinwise, not proof it works.

What it does

1. **Check in, in her own words.** On-device NLP turns "felt woozy" into answers she confirms.
2. **Review her medicines** against AGS Beers 2023, anticholinergic burden, fall-risk drugs and cascades; each finding is cited and ends as a question.
3. **See what changed.** Dizziness on **2 of 19** answered days before lorazepam, **9 of 14** after (Fisher's exact test). With no real change, 11.2% of simulated medicine changes still showed a large or clear difference (85.8% if symptoms come in runs of days, an assumed level).
4. **Bring one reviewed page to the visit,** plus a FHIR R4 export.

How it works

Cited KB → Engines → LLM rewords → Code checks → Human review

Deterministic TypeScript engines over a cited knowledge base (427 medicines, 89 Beers rules, 26 cascades) compute every fact. An optional on-device LLM (Qwen2.5, WebGPU) only rewords them; code drops sentences that fail a number, name or wording check.

Evidence & rigor

- **Clinical NLP:** micro-F1 **94.2%** (95% CI 90.7–97.1%) on 70 held-out stories never used for tuning; lexicon-only baseline 70.3%.
- **Medication review:** **57 of 57** synthetic cases pass (a written spec, not clinical validity); with one piece of knowledge removed, 33–54 of 57.
- **On-device AI:** 0 external requests in a recorded run; at least 6 flawed AI sentences (from 5 outputs) got through, so AI stays optional.
- **Tests:** 1,917 unit tests pass · 21 of 21 end-to-end checks pass · 46 axe scans, 0 violations.

Working

- Public live demo, no install
- NLP check-in · medicine matching
- Cited review · before/after · Ask Kinwise
- Reviewed brief · print · FHIR R4 export

Partial

- On-device LLM: optional, WebGPU only
- KB: 427 medicines, no clinician review
- No dose or allergy checks · English only
- One device, no sync · no FHIR receiver

Planned (not built)

- Pharmacist KB review · validation pilot
- Encrypted family sharing and storage
- Pharmacy, portal import · voice · Spanish

Hackathon prototype: not FDA reviewed or cleared, not clinically validated, no HIPAA claim. Designed never to say a medicine caused a symptom or to advise a change; the clinician decides. Emergency: call 911.

[1] 42 CFR 410.15 (detecting cognitive impairment). [2] Part D Comprehensive Medication Review, when the patient cannot take part because of cognitive impairment: 42 CFR 423.153(d)(1)(vii)(B)(2). [3] Tannenbaum et al., JAMA Intern Med 2014. [4] Vendor pages, Oct 2026; "not documented" is not proof of absence.